

Rupjyoti Barman

rupjyoti921@gmail.com | +91-9101171366 | linkedin.com/in/rupjyoti-barman-dev | github.com/rupjyotibarman

Summary

.NET Backend Developer with **3 years** of experience building **enterprise internal web platforms** using **ASP.NET Core** (Web API, MVC, Razor Views), **Entity Framework Core**, and **Role-Based Access Control (RBAC)** for Hewlett-Packard. Skilled in **C#, Clean Architecture, Factory Pattern, MSTest, and Moq**, with hands-on experience in **hardware diagnostic APIs, managed-unmanaged interop (P/Invoke, Native C++ DLL), WMI-based monitoring, and Windows System APIs (WUA, HPCMSL)** – consumed by **HP PC Hardware Diagnostics**. Experienced collaborating across global engineering teams in India, Taiwan, and the US.

Skills

- **Languages:** C#, C++, SQL, JavaScript, Python
- **Frameworks & Libraries:** ASP.NET Core (Web API, MVC), Entity Framework Core, WPF, Bootstrap, jQuery, LINQ
- **Windows APIs & Interop:** WMI, ManagementEventWatcher, WUA API, HPCMSL, P/Invoke
- **Architecture & Design:** SOLID Principles, OOP, Factory Pattern, Dependency Injection, Controller-Service-Repository Pattern
- **Database, DevOps & Tools:** SQL Server, PostgreSQL, EF Core (Code-First, Migrations), CI/CD, Azure Pipelines, Git, NuGet, Visual Studio, Postman, Jira, Confluence, Serilog
- **Testing & Security:** MSTest, Moq, JWT Authentication, RBAC

Experience

Software Developer, BoldTek India Pvt. Ltd. - Guwahati

June 2023 - Present

Client: Hewlett-Packard (HP)

- Engineered a **Defect Code module** (full CRUD, auto-incrementing error code generation, version history tracking) and a **Release Tracking Dashboard** (dynamic multi-entry JS-cloned form with bulk AJAX submission) within an internal **ASP.NET Core MVC** platform using **Razor Views, DTO/ViewModel** pattern, Controller-Service-Repository architecture with interface-based **dependency injection, EF Core Code-First**, and **SQL Server**, replacing manual Excel-based release documentation.
- Built a **MyIssues dashboard** with 3-tier **RBAC** (Viewer: read-only; ODM: leverage, test, and submit configurations for approval; Developer: create, edit with instant approval), Admin holding privilege to delete entries; implemented **client-side JS regex validation** enforcing key-value format, migrating **1000+ configuration entries** through the new validation logic; delivered UI enhancements using **Bootstrap and JavaScript**, with structured exception handling and **Serilog** logging.
- Led the India team in migrating from a **shared RPC-based Windows Service architecture** to a **decoupled three-layer NuGet-based architecture** (Business Logic, C# Interop Wrapper via P/Invoke, Native C++ DLL) with **Factory Pattern**-based dependency injection; achieved **100% elimination of third-party release delays** (previously 10–15 days) and reduced diagnostic development cycle from 4 to 3 sprints (40 to 30 working days, **25% reduction**) by removing 2 redundant architectural layers, resolving cross-team exceptions and sleep/hibernate-related service failures in **HP PC Hardware Diagnostics**.
- Developed **Business Logic and C# Interop Wrapper** (via P/Invoke) for **6 diagnostic modules** (Storage: DRV Read/Verify, SMART Health Monitor; Audio: Playback Test, Record Test; Network: NIC IRQ Validation, ROM/MAC Validation) to invoke native **C++ diagnostic APIs**, and independently built **2 modules from scratch** to detect and update **Drivers** (via **WUA API**) and **BIOS** (via **HP CMSL** for commercial systems), consolidating all into a centralized diagnostic backend with zero external service dependency.
- Replaced a service-based device discovery architecture (requiring a **2–3 minute service restart** per hardware change) by engineering a real-time **WMI monitoring module** (ManagementEventWatcher and WQL on Win32_PnPEntity) into the shared **Common** library; reduced detection latency **from 2–3 minutes to under 4 seconds (97% reduction)**, adopted across multiple hardware diagnostic APIs.
- Configured **Azure Pipelines** to trigger **NuGet package builds**, publishing the Business Logic package into the .NET Wrapper layer to produce the final consolidated NuGet package (bundling all layers including Native C++ DLL) for testing and client consumption.
- Verified diagnostic modules using a **WPF-based API testing application** and **MSTest** unit test projects across Business Logic and Wrapper layers, leveraging **Moq** for dependency mocking to ensure reliability prior to client integration.
- Partnered with cross-functional global teams (**India, Taiwan, US**) to deliver features, debug production issues, and align on diagnostic API design across release cycles.

Education

B.Tech in Computer Science and Engineering - CGPA: 8.01/10

May 2023

North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh